

Spatial relationship of city branding strategy adoption in megacity regions: Patterns and influencing factors

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ARTICLE INFO

Keywords:

City branding strategy adoption
Spatial relationships
Megacity regions
Influencing factors

ABSTRACT

With the rapid economic growth of regions, cities have adopted various city branding strategies in competitive and occasionally cooperative relationships with other cities. Intercity connections or regional context influence city branding strategy adoption, a topic that has been neglected by previous research on city branding. The spatial relationships of city branding strategy adoption is a worthy topic due to preliminary signs of the relevance of such connections or relationships. This paper explores the spatial relationship of city branding strategy adoption in megacity regions from two perspectives: patterns and its influencing factors. Regarding patterns, megacities regions have shown spatial relationships in different city branding strategy adoption in both within and cross-regional comparisons. Considering influencing factors, this study finds that city branding strategy adoption in the three megacity regions has spatial correlations by using the spatial analysis method, while city branding strategy adoption in the same region tend to be influenced by economic or urban factors in similar ways. Thus, policymakers need to consider these spatial relationships in city branding designs and pay attention to the factors influencing city branding strategy adoption in their own regions.

1. Introduction

Worldwide, cities have to face severe competition during the urban development process, and they tend to use various strategies to brand themselves. City branding strategies can range from advertising and promotion to improve the exposure and reputation of a city, to policies that attract specific industries or particular groups of visitors or residents (Lu, de Jong, et al., 2020), to more comprehensive and specifically designed urban governance strategies that state the vision and outline developmental pathways of local governments (Ye & Björner, 2018). For example, attractive slogans and city logos are used in city images. Cities engage in building international reputations by organised mega events, such as the Qatar World Cup (Brannagan & Giulianotti, 2015). These strategies can be classified into three types: city promotion, city marketing and city branding (Ma et al., 2021).

City branding strategies are not adopted in a vacuum; they are used within their own spatial relationships in regional contexts. Cities can compete with or learn from other cities in the same region. In past decades, megacity regions have played a greater role in the urbanisation process (Yeh & Chen, 2020). They are comprised of clusters of people, resources and industries in a large urban area with a few core cities. As

both competition and cooperation exist in megacity regions, the city branding strategies they adopt exert influence over other cities (Xy & Yeh, 2005). Thus, spatial relationships can exist in city branding strategy adoption, which is visible in the similarity of branding means, channels, target groups and branding aims in the regional context.

Several studies have attempted to explore the spatial relationship between individual city branding strategies in the regional context. De Jong et al. (2018) explained city branding adoption in terms of regional importance and industrial upgrading processes. Lu and de Jong (2019) further explained the city branding evolution that came from intervention from higher levels of government, the economic situation, and city imitation behaviours in the megacity regions of China. However, these studies focused on branding content rather than on sophisticated city branding strategies. Recently, Ma et al. (2021) have distinguished city promotion, city marketing and city branding strategies based on the goals and characteristics of Guangdong province and two special administrative regions in China. Their research is one of the first attempts to explore city branding strategy adoption in a particular region. However, the literature has neglected to review the spatial relationship patterns and their influencing factors of city branding strategy adoption in the regional context.

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<https://doi.org/10.1016/j.cities.2023.104567>

Received 19 March 2023; Received in revised form 2 September 2023; Accepted 16 September 2023

Available online 25 September 2023

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Against the background of megacity region development at the global level, many cities are putting efforts into establishing and promoting their brands in the regional context, especially in developing countries (Chatterjee & Chattopadhyay, 2020). The spatial relationship patterns and their influencing factors of city branding strategy adoption in megacity regions are also important lessons that help these cities to tailor their branding strategies. Due to limited academic attention has been paid to these perspectives, the aim of this research is to explain the spatial relationship patterns and their influencing factors of city branding strategy adoption in megacity regions.

This research uses three megacity regions in China as a case study. Megacity regions in China emerged in the late 1970s and played a significant role in the progress of urbanisation and economic development. In 2019, these three megacity regions in China occupied nearly 5 % of the land, but generated 40 % of the national GDP and housed >63 % of the national population (Yeh & Chen, 2020). The spatial relationship patterns and their influencing factors of city branding strategies in these three megacity regions can be used to understand city branding strategy adoption in the regional context.

The structure of this paper is as follows: Section 2 reviews previous studies about spatial relationship patterns and their influencing factors of city branding strategy adoption in megacity regions. Section 3 introduces the data collection and analysis process. Section 4 presents the empirical results. Section 5 discusses the key findings. Section 6 concludes with policy implications, research limitations, and outlooks.

2. Literature review

2.1. City branding strategy adoption and spatial relationships

City branding strategies are the branding or marketing measures applied in urban governance and place management (Ma et al., 2021). City branding strategy as used here is an umbrella concept referring to cities' paradigmatic approaches to the management of their symbolic capital in the pursuit of local (economic) development (Dinnie, 2010). City branding strategies develop from promotional to more manageable and sophisticated marketing, followed by more deeply symbolic anchored branding (Anholt, 2010). With the development of Internet technology, especially social media, citizens, enterprises, and other stakeholders can also participate in the city branding generating process, and the difference between city brand creators and receivers become more insignificant (Lucarelli, 2019; Vallaster et al., 2018). However, municipal governments tend to have a strong will to control their city brands to influence their targeted audience (specially in the multi-level governance in the Chinese context) (Lu et al., 2017). Thus, city branding strategy adoption in this research narrowly refers to city branding promoted by government actions rather than the broad concept of city branding co-creation. The common classification of city branding strategies includes city promotion, city marketing, and city branding (Boisen et al., 2018; Ma et al., 2021). These city branding strategies can be understood and distinguished according to their

definitions, goals, policy measures and target groups (See Table 1).

Besides the classification of city branding strategy, scholars also explore the adoption of city branding strategies, and one often neglected factor is their spatial relationships in regions. In the place branding literature, scholars have already started to emphasise the geographic features of place branding (Pike, 2009). The spatial relationships between city branding strategies in regions, as well as their influencing factors, have been touched upon in a few studies. In the Negev region in Israel, Aharon and Alfasi (2022) found that each place is affected by at least one other place in the region, and that the geographical origins of cities shape city brands in ways that make these brands difficult to alter. In the Pearl River Delta in China, Lu and de Jong (2019) found that cities tend to take cues from the core cities of their regions when they attempt to catch up with other cities' branding activities.

Based on the previous literature, there are at least two streams of studies explaining spatial relationships in the adoption of city branding strategies. The first stream unfolds the spatial relationships of city branding strategies through a regional governance lens based on a hierarchy perspective (Boisen et al., 2011; Herstein, 2012; Lu et al., 2017). As important governance bodies in their regions, some regional governments care about city branding strategies at the local level. They can influence city branding channels, contents and standards to some extent (Zenker & Jacobsen, 2015). For instance, the Chinese national government launched a national examination of city websites based on their effectiveness and performance, which promote the standardisation of city branding strategy based on city websites (SC, 2015). Additionally, regional governments can influence how cities brand themselves through urban design, architecture style and other elements (de Jong & Lu, 2022). Ochkovskaya and Gerasimenko (2018) found that new political powers can influence how municipal governments promote themselves through urban design. Thus, city branding strategies can have spatial relationships when regional governments pay attention to or even intervene in city branding strategy adoption.

Another way to explain the spatial relationships between city branding strategies focuses mainly on the city actors on the horizontal level, namely city competition or imitation behaviours (Xu & Yeh, 2005). One typical example of city competition behaviours is city marketing approaches, as cities see each other as competitors in attracting their target groups in the region. Cities try to differentiate themselves from each other to attract tourists, enterprises and talents (Kavaratzis, 2007). In addition to city competition, city imitation behaviours are more often found among small cities or cities with peripheral locations. These cities do not have unique city brand identities, and they tend to be influenced by larger and more prominent cities in the same region (Niedomysl & Jonasson, 2012). Thus, city imitation or competition behaviours influence the spatial relationships in city branding strategy adoption in different ways.

Above all, previous scholars have investigated spatial relationships in city branding strategies and tried to explain from regional governance and city interaction behaviour perspectives. As city branding strategies also aim to solve the problems they face in urban development, this

Table 1
Classification of city branding strategies.

	City promotion (CP)	City marketing (CM)	City branding (CB)
Definition	Consciously promoting cities by communicating selective images to a target market (Cudny, 2019)	Activities or policy instruments that attract various target groups to a city for the long term (van den Berg et al., 2016)	Helping cities to compete with other cities at the regional or international level (Kavaratzis, 2004)
Goals	Creating an attractive image and convincing people to come and visit (Dong et al., 2014)	Identifying and responding to the needs of target groups (Merrilees et al., 2012)	Urban transformation or city development (Ma, 2021; Yang et al., 2018; Zenker et al., 2017)
Policy measures	Advertising, media activities (Avraham, 2004)	Preferential policies to attract investment, talents or companies (Udir Mišić & Podnar, 2019)	A whole set of actions that build a positive reputation for the city (Vanolo, 2008)
Target groups	No distinctions between specific groups (Ma et al., 2021)	Distinctions between various target groups (Ashworth & Voogd, 1990; Merrilees et al., 2012)	More target groups are identified, such as local governments, cultural and sports organisations, promotional agencies, businesses, academic institutions, residents, and tourists (Stubbs & Warnaby, 2015)

research further explores influencing factors of spatial relationships in city branding strategy adoption based on economic and urban factors.

2.2. Factors influencing city branding strategy adoption

Besides regional governance and city interaction, the spatial relationships of city branding strategy adoption can also originate from urban transition problems (Prilenska, 2012). Essentially, cities dealing with urban transition challenges will adopt city branding strategies influenced by their economic and urban factors (Yang et al., 2018). Referring to previous research, many scholars have proposed other factors that play into adoption decisions, such as economic levels and industrial structures (Dudek-Mańkowska & Grochowski, 2019; Zhang et al., 2020). Various urban factors can also influence city branding strategies, such as population scale (Yang et al., 2018), regional importance (Bonakdar & Audirac, 2020), or the city's competitiveness (Anttiroiko, 2015). Some of these studies have mentioned the population scale, which can be a crucial urban factor (Burnell & Galster, 1992; Rozenfeld et al., 2011). The following paragraphs elaborate on the influencing factors.

First, cities are economic growth engines, and the economic development stage of cities can influence their branding strategy adoption (Ma et al., 2020). GDP per capita is one way to measure the economic status of cities (Moomaw & Shatter, 1996). As a rule, cities with higher GDP per capita have more opportunities to utilise different promotional channels, attract various marketing targets and organise relevant branding activities than those with lower GDP per capita (Lu, Ma, et al., 2020). When cities have stronger economic capacities, they tend to adopt more sophisticated branding strategies.

Second, cities with diverse industrial structures can have different city branding strategies. Cities with agriculture- and extraction-oriented sectors may be less concerned about city marketing and branding, as they face considerable pressure to attract target groups or address urban transition problems (de Jong et al., 2018). Cities with manufacturing and production-oriented sectors or mainly trade and service-oriented sectors may spend more time on city marketing and branding, since they pay more attention to attract target groups and use branding as a tool for transitions (Ma et al., 2019; Vanolo, 2008).

Third, cities with large populations may wish to attract more attention from outside than small or medium-sized cities (Rozenfeld et al., 2011). The population scale of a city impacts its importance in a region, as cities are believed to support current economic activities and attract new resources from the global value flows (Cramer et al., 2004). Large populations can lead to higher income levels, more occupational choices, and greater availability of goods and services, which demand a larger population threshold (Burnell & Galster, 1992). Therefore, city branding strategies in large cities can be rather different from those in smaller ones.

2.3. Spatial relationships of city branding strategy adoption in megacity regions

The rise of such megacity regions can be witnessed around the world, and popular cases can be found in developed countries, such as South East England, the Randstad region, and the Rhine-Ruhr region (Goess et al., 2016; Hall & Pain, 2006). Similarly, in the past two decades, the emergence of megacity regions can also be observed in Asia and Latin America (Chatterjee & Chattopadhyay, 2020). In China, promoting the development of megacity regions has been an essential urbanisation strategy in recent years (Luo & Shen, 2009; Yang & Li, 2013). Under this trend, regional governments in megacity regions have issued spatial plans to guide regional development (Lucarelli & Heldt Cassel, 2020). However, it is difficult to implement spatial planning in many places due to vague and abstract planning strategies. Thus, city branding has become a valuable addition to the toolbox of planners in the regional development process (van Assche et al., 2020).

In the city branding in megacity region literature, one stream of studies focuses on branding content. For instance, Goess et al. (2016) found differences in the branding content of cities between the Dutch Randstad region and the German Rhine-Ruhr region. The more industrialised Ruhr region focuses on green, liveable, and knowledge-oriented brands, while cities with stronger knowledge-intensive sectors in the Randstad or Rhine regions portray themselves as smart or sustainable. Han et al. (2018) also compared the city branding content of the Northeastern Region with the leading three megacity regions in China. They found that branding content of cities in the Northeastern Region tends to match the reality and not as ambitious as those of the leading Chinese regions.

Another approach to study city branding in megacity regions focuses on the influencing factors of city branding. Using cities in Guangdong province as an example, Ma et al. (2021) found that economic factors positively influenced the adoption of city branding, marketing and branding strategies, whereas urban size was positively associated with these city branding strategies. Additionally, as cities are embedded in multilevel governance in some institutional contexts, it is necessary to consider the impact of higher level governments on city branding practices. Thus, generally speaking, in countries with stronger hierarchical governance, higher level governments play an important role in influencing the content of city branding (Lu, de Jong, et al., 2020; Ye & Björner, 2018).

These studies of city branding content and influencing factors have shown that city branding strategies may have spatial relationships, but less attention has been paid to them. One attempt to study the spatial relationships between copycat behaviours of cities in their branding practices was based on the sample in Pearl River Delta region (Lu et al., 2017). However, the research is still limited to descriptive method instead of quantitative analysis. Due to the growing number of cities in megacity regions, it is possible to adopt spatial regression methods to explore the spatial relationships between city branding strategy adoption in megacity regions further.

3. Research design

3.1. Data collection

The three megacity regions used in this study are Jing-Jin-Ji (JJJ), the Yangtze River Delta (YRD) and the Greater Bay Area (GBA), as illustrated in Fig. 1. JJJ comprises 11 cities including Beijing, Tianjin, and cities in Hebei province. The YRD has 41 cities covering Shanghai and cities in Jiangsu, Zhejiang, and Anhui provinces. The GBA has Hong Kong, Macau, and nine other cities on the mainland. In this research, we also include 12 other cities in Guangdong province to increase the GBA sample. Thus, the GBA represents the cities in Guangdong province and two special administrative regions, Hong Kong, and Macau. The research sample includes 77 cities in total.

The CP, CM and CB strategies are assessed in the following ways. Ma et al. (2021) formulated an analytical framework for a city branding strategy adoption based on their differences in definitions, policy measures and target groups. We applied and adjusted this framework. In CP strategies, we deleted digital brochure, as most Chinese cities do not have them yet. In CB strategies, although stakeholder involvement plays an important role in CP strategy adoption, we deleted the corresponding indicators in city branding processes because secondary data of stakeholder involvement in all 77 cities are difficult to collect. Finally, the indicator framework of city branding strategies is outlined in Fig. 2.

Appendix Tables A1-A3 shows the framework of indicators we used for the three city branding strategy concepts and presents the data sources. The contents of the official government websites for each city provided information for all 77 cities we studied (indicators CP1-4 and CB1-3). Descriptions for city identities and visions were collected from the Urban Master Plans of each city (indicator CB4). We chose the latest version of Urban Master Plans we can find when writing this manuscript.

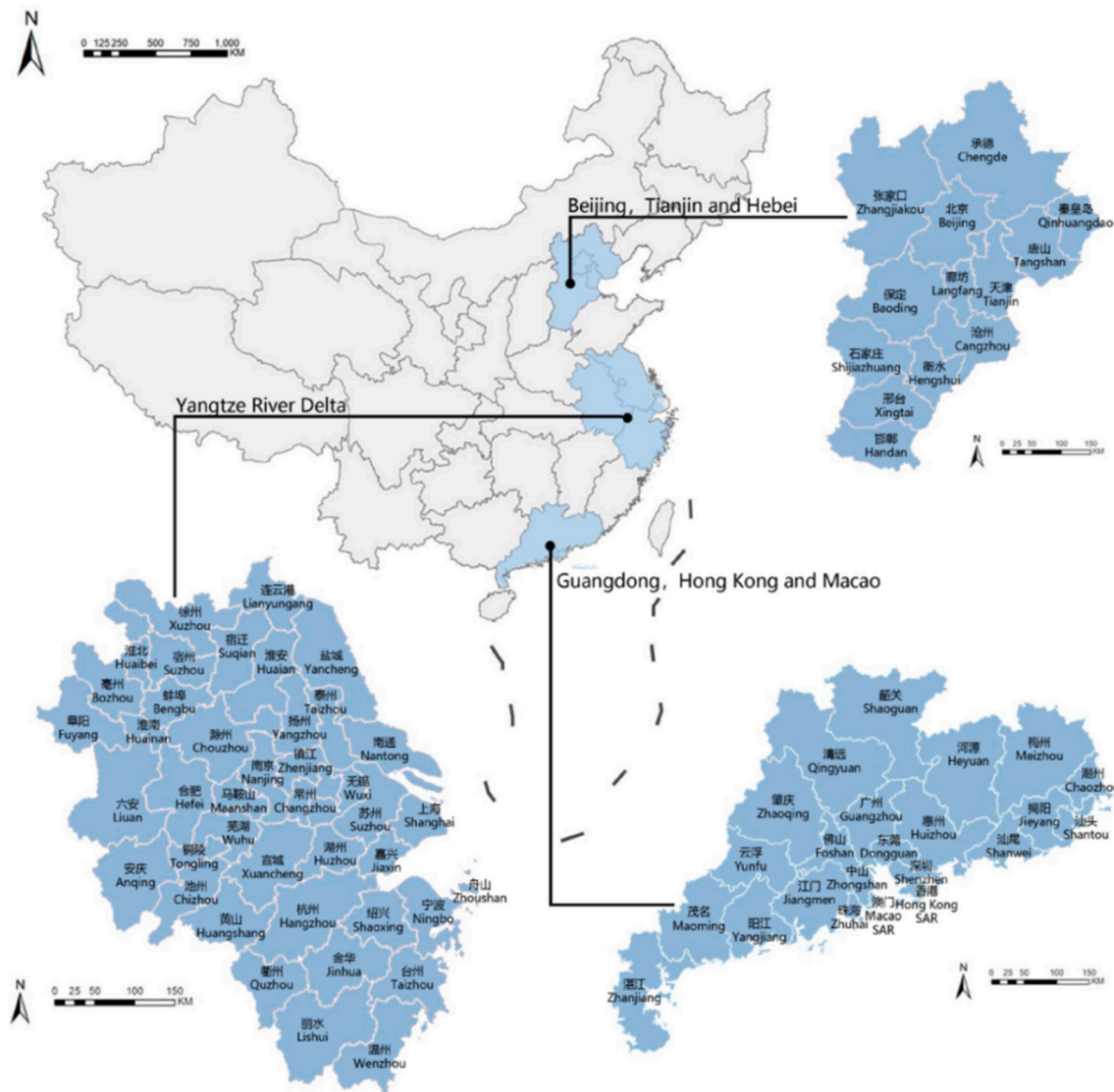


Fig. 1. Location of three megacity regions in China (Li et al., 2022).

The specific policies and projects undertaken by the cities in the context of city branding (Tables A2 and A3, indicators CM1–3 and CB5) were identified via an analysis of government work reports. Government work reports (from 2016 to 2020) are the appropriate source for city branding-related policies and projects, as municipal governments in China need to summarise their government activities annually through these reports.

3.2. Data analysis

In the following three steps (Fig. 3), we analysed the spatial relationships of city branding strategy adoption in three megacity regions. First, regarding spatial relationships within regions, we presented the average value and standard deviation of CP, CM and CB strategy scores within each megacity region. We also ranked the CP, CM and CB strategy scores of cities in each megacity region. Second, concerning the spatial relationships between city branding strategies across megacity regions, we calculated the z -scores of CP, CM and CB strategies to ensure that they were comparable. Then, we listed the top and bottom five cities according to the city branding strategy score of CP, CM and CB. To investigate the mix strategy of city branding strategies, we further compared the standard deviations within CP, CM and CB strategy scores

of each city. Finally, we explored the factors influencing city branding strategy adoption in the three regions using Moran's I test and a geography weighted regression (GWR) model.

Specifically, as the dependent variables CP, CM and CB are aggregated scores with spatial autocorrelation, we adopted the GWR model to explore the influencing factors of city branding strategy adoption in three regions. GWR constructs a separate OLS equation for every location in the dataset, which incorporates the dependent and explanatory variables of areas that fall within the bandwidth of each target location (Cardozo et al., 2012). Explanatory variables consist of indicators from economic, urban, and other factors (Table 2). The variables within the economic factor include GDP per capita and industrial structure. The urban factors in this research refer to population scale. We used the average number from 2016 to 2020 to calculate the number of each explanatory variable due to data availability and consistency. Specifically, we chose GDP per capita (GDP) to represent the economic development level of a city. Tertiary industrial sector (Tertiary) refers to the proportion of tertiary industries in GDP (Zeng et al., 2019). We employed the permanent population (Population) as a proxy for the population scale of cities (Lu, Ma, et al., 2020).

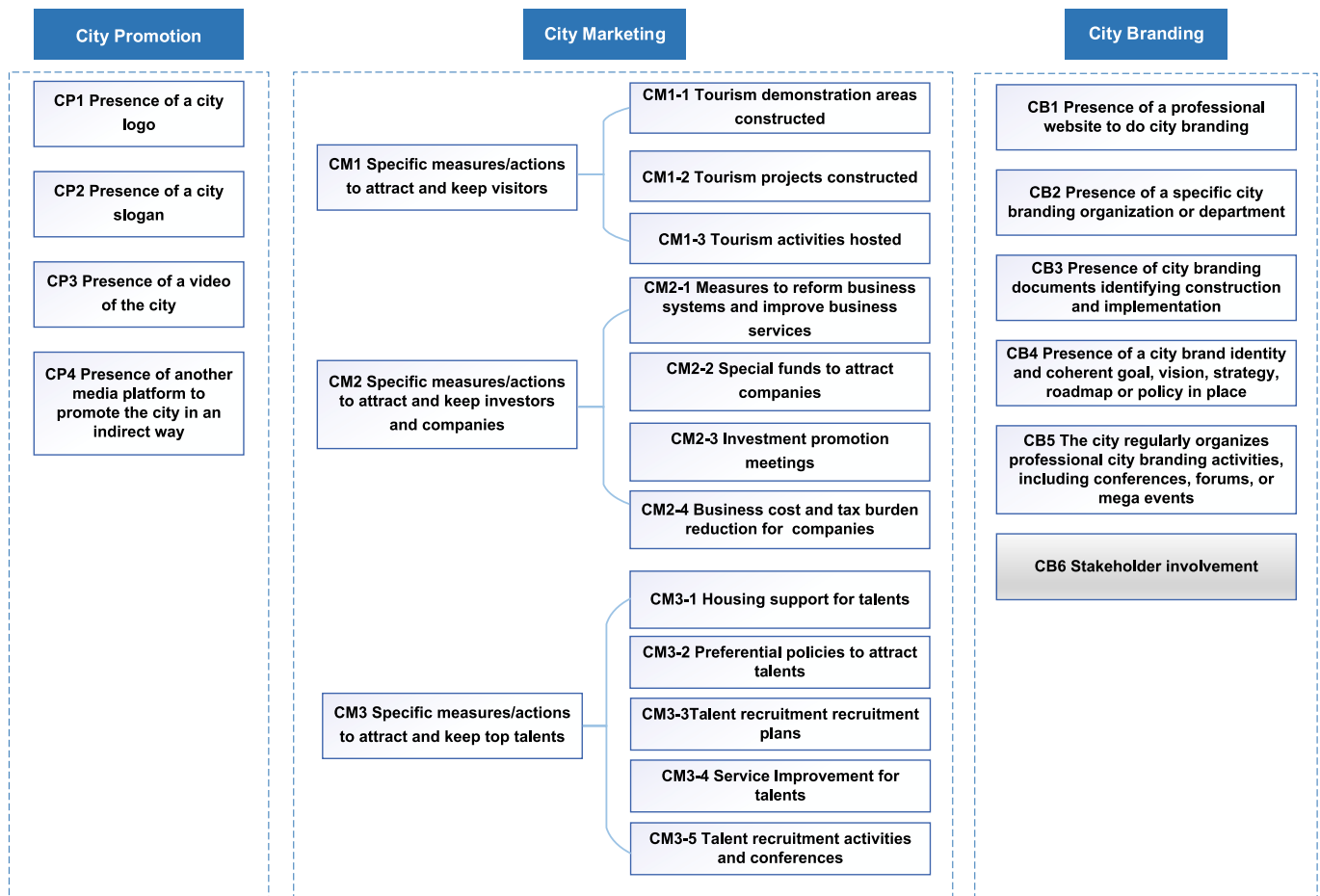


Fig. 2. Indicator framework of city branding strategies (adjusted based on Ma et al., 2021). The indicators in grey indicates that data are not available in this research).

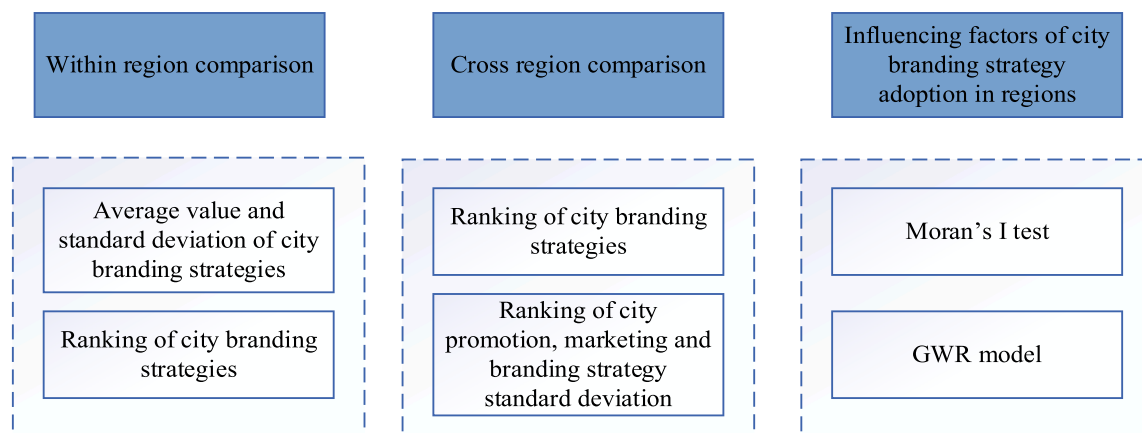


Fig. 3. Analysis process for spatial relationships and influencing factors of city branding strategy adoption.

4. Empirical results

4.1. Spatial relationships of city branding strategy adoption within regions

First, regarding the average value, GBA cities had the highest CP and CB strategy scores, as Table 3 shows. YRD has the highest score in the CM strategy. JJJ does not have prominent city branding strategies compared with the other two regions. Regarding the variances of three regional strategies in CP, CM and CB, we find that the variances of the

three strategies in the YRD and GBA are greater than those of JJJ. Meanwhile, the variance of CB in the YRD is higher than that of the other indicators. The variance of CM in the GBA is more significant than in other regions, which means the YRD and GBA have both leading and lagging cities in terms of CB and CM strategies.

Second, we further rank each region's CP, CM and CB strategies according to their scores. Regarding CP strategies (Fig. 4), Tianjin and Shijiazhuang rank in the top two as regards CP strategies in JJJ, while Beijing only ranks seventh among fourteen cities in JJJ. Beijing does not

Table 2
Descriptive information of variables.

Variable	Obs	Unit	Mean	Std. Dev.	Min	Max
CP	77	score	2.091	0.809	0.990	4.950
CM	77	score	9.270	2.032	5.610	15.510
CB	77	score	4.668	1.687	1.850	13.120
Population	77	million	5.575	3.143	0.666	16.648
GDP	77	thousand RMB	94.769	73.501	28.550	563.836
Tertiary	77	%	0.428	0.106	0.311	0.929

Table 3
Average value and standard deviation of CP, CM and CB strategies.

	Average value			Standard deviation		
	JJJ	YRD	GBA	JJJ	YRD	GBA
CP	1.485	2.108	2.241	0.764	0.659	1.189
CM	8.860	9.875	8.650	0.938	2.013	2.259
CB	4.570	4.239	5.490	0.647	1.196	1.168

have a special design for its website, and it only provides one city promotional video. The various access options are also not directly shown on this website. Regarding the YRD, the scores in YRD cities have a flatter trend, which means the cities have similar CP strategies. Regarding the GBA, Hong Kong and Zhuhai have higher scores among GBA cities. Hong Kong has a special website for CP with abundant city promotional videos. Hong Kong also provides several options for international access, including YouTube, Twitter, etc. In mainland cities in Guangdong, Zhuhai also has a special design for its website with a variety of city promotional videos. This website also provides WeChat, Weibo, and Mobile app access.

Regarding CM strategies, in JJJ, Beijing and Shijiazhuang have good performance in CM, but other cities lag far behind (Fig. 5). In JJJ, the

CM strategy scores of Beijing and Tianjin are significantly higher than those of cities in Hebei province. CM strategy scores among cities in Hebei province do not show a large variation. In the YRD, Nanjing, Shanghai and Hangzhou have actively implemented CM strategies. Compared with JJJ, the YRD has paid more attention to improving the business environment. Thus, the region can attract more talented people because of its large number of universities, higher education institutions, and foreign-funded enterprises. The scores in GBA cities have a flatter trend, which means they have similar CM strategies. Still, the scores of Shenzhen, Dongguan, Zhongshan, Foshan and Guangzhou are higher than others in Guangdong province. This can be explained by the fact that these cities have adopted more preferential policies to attract investment and talent, and actively to improve their business environments.

Regarding CB strategies, the average scores of JJJ and the YRD are not high (Fig. 6) and there is little difference in terms of CB strategy adoption within these two regions. Concerning the GBA, the scores of CB strategies of cities show significant differences. Specifically, Hong Kong, Guangzhou and Shenzhen obtain top scores, while other cities in the mainland are lagging behind. Compared with mainland cities, Hong Kong has more experience with CB strategies, as it has committed to constructing and promoting city branding for many years. Hong Kong has positioned itself as “Asia’s World City” since 1999, and it has a professional organisation and website for its city branding. In contrast, most cities in JJJ and the YRD do not yet have professional city branding organisations or websites.

4.2. Spatial relationships of city branding strategy adoption cross regions

Concerning spatial relationships in cross-region comparisons, we further compared cities’ CP, CM and CB strategy scores. We calculated the z-score of CP, CM and CB strategy scores to ensure that they were comparable. First, concerning city ranking, Table 4 presents the top and the bottom five cities. We found that the top five cities in CP, CM and CB

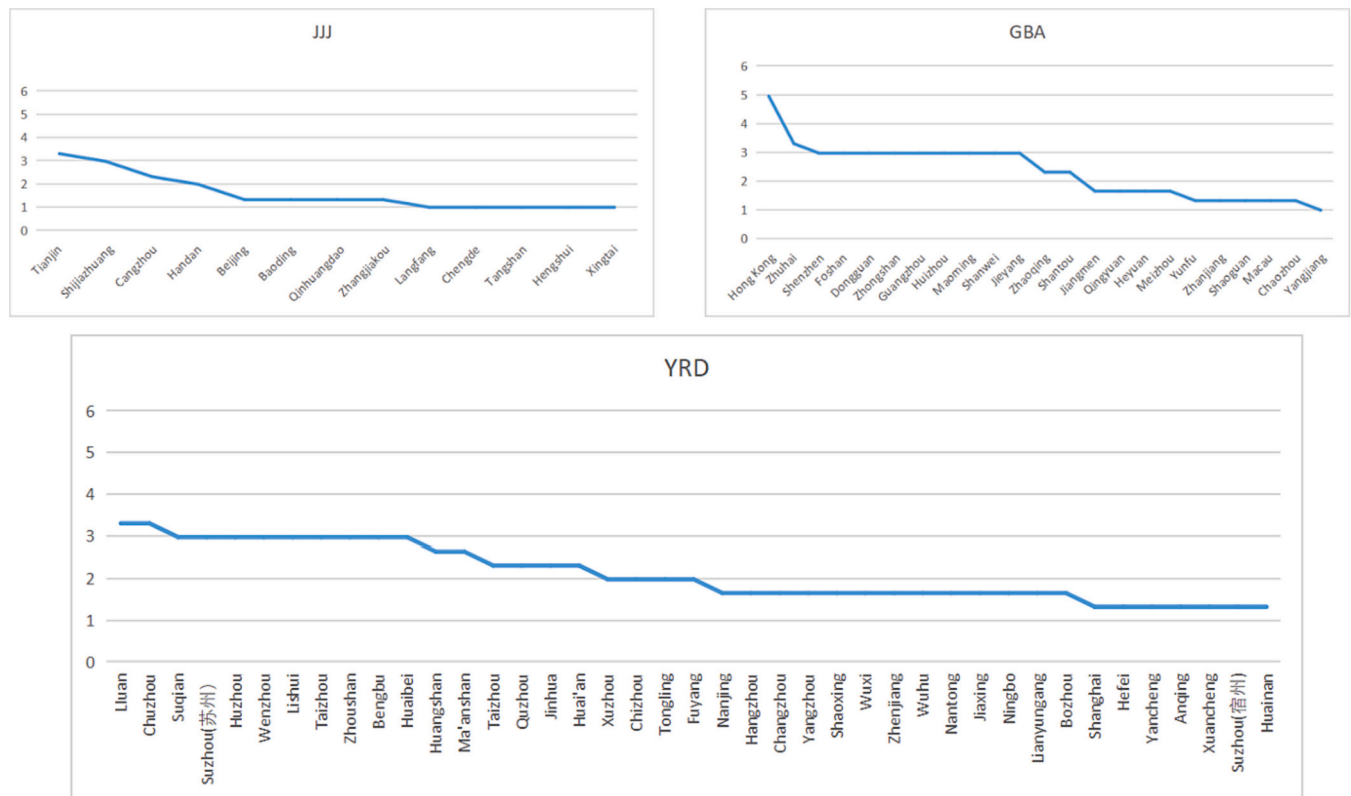


Fig. 4. City promotion strategies of JJJ, GBA and YRD.

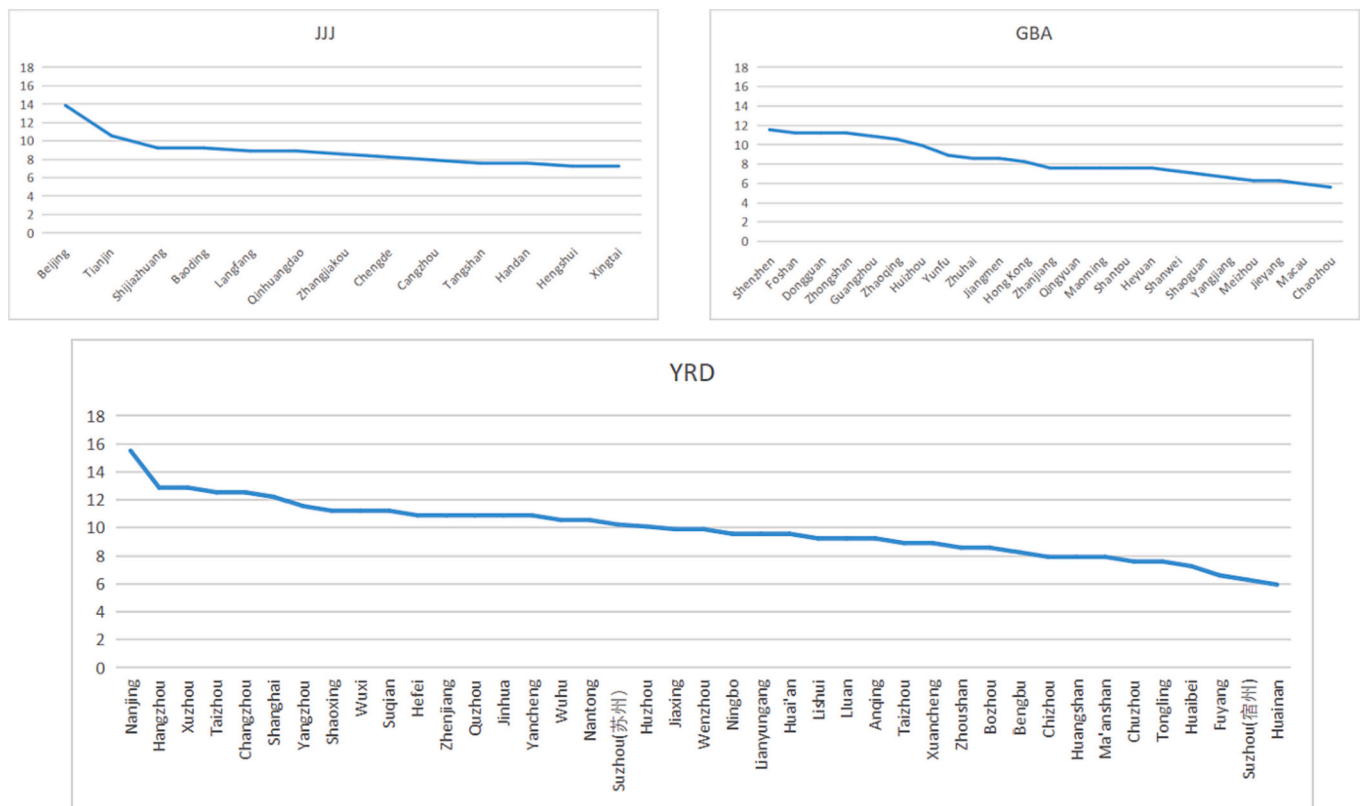


Fig. 5. City marketing strategies of JJJ, GBA and YRD.

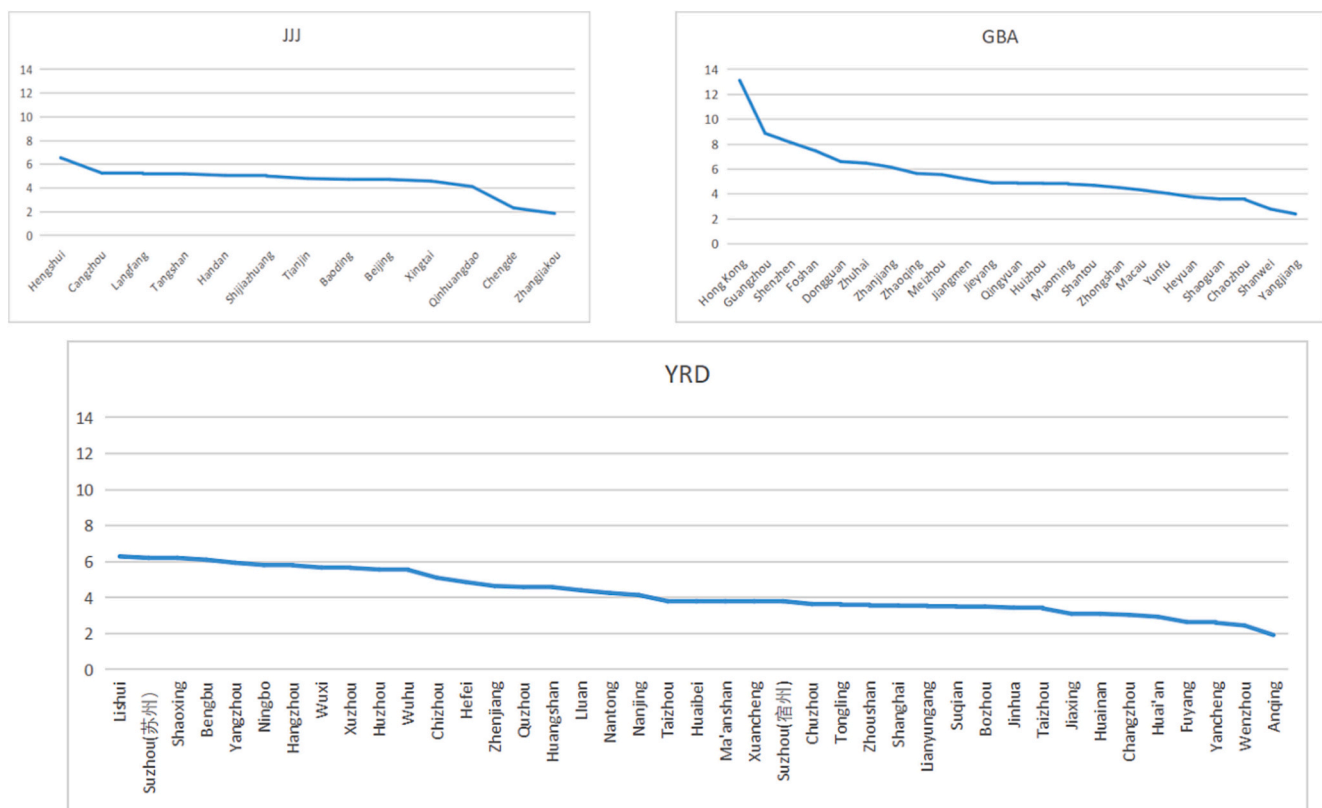


Fig. 6. City branding strategy adoption of JJJ, GBA and YRD.

Table 4

The top and bottom five cities according to their city branding strategy scores.

	City	Region	CP	City	Region	CM	City	Region	CB
Top five	Hong Kong	GBA	4.950	Nanjing	YRD	15.510	Hong Kong	GBA	13.120
	Tianjin	JJJ	3.300	Beijing	JJJ	13.860	Guangzhou	GBA	8.880
	Liuan	YRD	3.300	Hangzhou	YRD	12.870	Shenzhen	GBA	8.150
	Zhuhai	GBA	3.300	Xuzhou	YRD	12.870	Foshan	GBA	7.466
	Chuzhou	YRD	3.300	Changzhou	YRD	12.540	Dongguan	GBA	6.602
Bottom five	Xingtai	JJJ	0.990	Chaozhou	GBA	5.610	Zhangjiakou	JJJ	1.850
	Hengshui	JJJ	0.990	Huainan	YRD	5.940	Anqing	YRD	1.920
	Tangshan	JJJ	0.990	Macau	GBA	5.940	Chengde	JJJ	2.320
	Chengde	JJJ	0.990	Jieyang	GBA	6.270	Yangjiang	GBA	2.400
	Langfang	JJJ	0.990	Meizhou	GBA	6.270	Wenzhou	YRD	2.450

are all in the YRD or GBA, except Beijing and Tianjin. The bottom five cities in the three strategies are all small cities, and a large proportion of them are in JJJ. Above all, these results show that leading and lagging cities that employ city branding strategies tend to be in specific regions.

Second, we calculate the standard deviation within the CP, CM and CB strategy scores of individual cities, which can show the mix strategy of their city branding strategy adoption (see Table 5). We found that the cities with the largest deviations in z-scores included Hong Kong, Nanjing, Beijing, Jieyang and Shanghai. Among these five cities, only Jieyang is not a first-tier city. This indicates that these first-tier cities focus on one or two city branding strategies, rather than reaching a balance among the three strategies. Specifically, Hong Kong has a relatively high score in both CP and CB strategies but lower scores in CM strategies. Interestingly, Beijing, Nanjing and Shanghai all have relative high scores in CM strategies but lower scores in CP and CB strategies. This indicates that the first-tier cities on the mainland have focused more on CM strategies, while CP and CB strategies are still lagging compared with those of Hong Kong. For instance, most mainland cities do not have their own professional city branding organisations or websites.

The cities with the smallest standard deviations include three cities from the GBA and two from the YRD. Only Shenzhen is a first-tier city, and its z-scores in CP, CM and CB strategies are all positive. The other four cities have negative scores in one or even three city branding strategies. This means that Shenzhen has paid attention to all three city branding strategies, while the other four cities do not perform well in all three city branding strategies. Since Shenzhen lies adjacent to Hong Kong, it has learned from Hong Kong's CP and CB strategies. The other cities, both first-tier and small and medium-sized cities, have the potential to reach a balance among the three city branding strategies in the future.

4.3. Factors influencing city branding strategy adoption

Since city branding strategies have regional features, we further explored the factors influencing city branding strategy adoption in the three regions. We used Moran's *I* test to check for spatial autocorrelation in dependent variables with Knn-4 spatial weights (Table 6). The test results showed very small *p*-values for city branding strategy scores,

Table 5

Z-score and standard deviation of CP, CM and CB strategy scores in top and bottom five cities.

	City	Region	Z_CP	Z_CM	Z_CB
Top five	Hong Kong	GBA	3.543	-0.512	5.036
	Nanjing	YRD	-0.531	3.064	-0.311
	Beijing	JJJ	-0.938	2.251	0.036
	Jieyang	GBA	1.099	-1.488	0.140
	Shanghai	YRD	-0.938	1.438	-0.661
Bottom five	Shenzhen	GBA	1.099	1.113	2.077
	Yangjiang	GBA	-1.346	-1.325	-1.345
	Zhanjiang	GBA	-0.938	-0.837	0.887
	Huai'an	YRD	0.284	0.138	-1.027
	Suqian	YRD	1.099	0.951	-0.690

Table 6Moran's *I* tests for spatial autocorrelation of independent variables.

		Moran's <i>I</i>	<i>P</i> -value
JJJ	CP	-0.168	0.305
	CM	0.079	0.085
	CB	0.132	0.078
YRD	CP	-0.127	0.122
	CM	0.099	0.097
	CB	-0.137	0.139
GBA	CP	0.181	0.045
	CM	0.379	0.001
	CB	0.237	0.020

indicating the existence of spatial autocorrelation. This dictated that we should use a GWR approach to explore the influencing factors of city branding strategy adoption.

Table 7 shows the impact of influencing factors on CP, CM and CB strategy scores. In the JJJ model, it shows that cities with larger populations and higher tertiary proportions in industrial structures tend to have higher CM scores. This means that larger cities or cities with higher tertiary sectors tend to market themselves to attract tourists, businesses or talents. Another observation is that no influencing factors could be found in the CP and CB models.

In the YRD models, cities with higher GDP per capita tend to have higher scores in CM and CB strategies. In the CM model, in addition to the economic development level, cities with larger populations tend to pay more attention to marketing in tourist, business or talent attractions. In the GBA model, cities with larger populations tend to have higher scores in CP, CM and CB strategies. The results indicate that cities with larger populations in the GBA also have the awareness and willingness to adopt city branding strategies.

As the GWR model can provide the differentiated coefficients of explained variables, it is possible to analyse the various impact of influencing factors on city branding strategy adoption in the three regions. The geographical distribution of the population in the CM model in JJJ and the YRD, as well as that of the population in the CB model in the GBA, are in Fig. 7. These results show that cities in the same region tend to have their own influencing factors. In Fig. 7, it is also found that the range of coefficient is small (e.g. coefficient of population ranges from 0.1081 to 0.1084 in CM model in JJJ), which indicates the influential intensity of these influential factors is similar within their regions.

5. Discussion

In the spatial relationship of city branding strategy adoption within regions, GBA cities have the highest average value of CP and CB strategies, and the YRD has the highest score in CM. The YRD and GBA have the largest variances of CB and CM, which implies that they have both leading and lagging cities in CB and CM strategies. City branding strategies in JJJ still remain at a less developed stage than the other two regions.

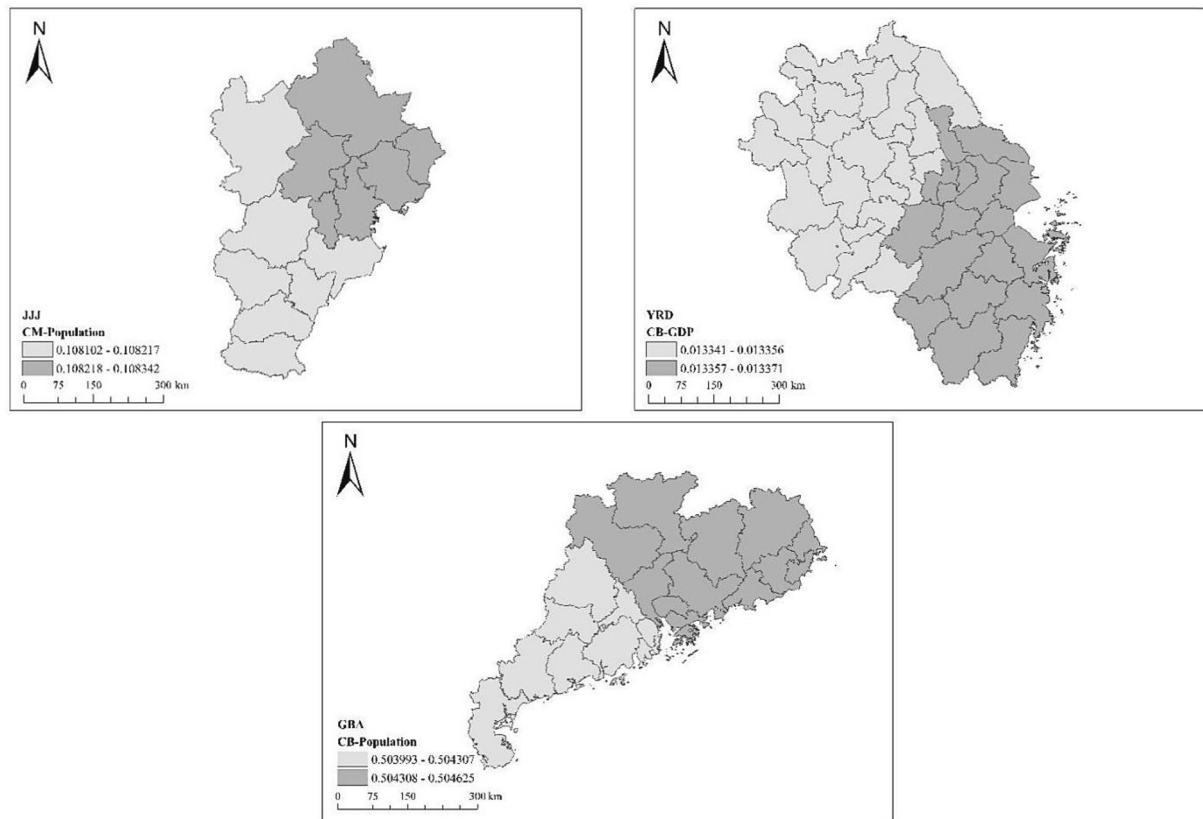
The variation in city branding strategy adoption in the three regions

Table 7

Influencing factors of city branding strategies in JJJ, the YRD and the GBA in the GWR model.

Variable	JJJ			YRD			GBA		
	CP	CM	CB	CP	CM	CB	CP	CM	CB
Population	0.092 (0.075)	0.108* (0.060)	0.111 (0.110)	−0.055 (0.038)	0.134 (0.094)	−0.093 (0.065)	0.143* (0.093)	0.373* (0.211)	0.504*** (0.158)
GDP	0.003 (0.009)	0.006 (0.007)	0.017 (0.013)	−0.002 (0.003)	0.027*** (0.007)	0.013*** (0.005)	0.0001 (0.005)	0.011 (0.010)	−0.0003 (0.008)
Tertiary	−0.897 (2.818)	11.576*** (2.246)	−5.983 (4.160)	−0.335 (2.223)	9.345* (5.344)	1.924 (3.801)	1.658 (3.672)	−9.529 (7.858)	8.034 (6.209)
Bandwidth	58.355	58.355	58.355	12.152	3.944	70.839	9.547	2.448	71.387
Residual Squares	6.008	3.817	13.091	15.339	67.181	45.513	14.996	38.703	43.998
Effective Number	4.009	4.009	4.009	4.454	7.927	4.014	4.398	8.395	4.007
Sigma	0.817	0.652	1.207	0.648	1.425	1.109	0.898	1.629	1.522
AICc	45.461	39.561	55.584	88.386	155.028	132.367	69.699	102.923	93.732
R ²	0.208**	0.899	0.288	0.118	0.586	0.206	0.259	0.517	0.625

Note: standard errors are in parentheses.

* $p < 0.1$.** $p < 0.05$.*** $p < 0.01$.**Fig. 7.** Coefficients of influencing factors of city branding strategies in JJJ, the YRD and the GBA.

reflects their different regional characteristics. For JJJ, the regional integration level is still low, and it only has Beijing and Tianjin as core cities. The economic development status of other cities in Hebei province is also lower than those of the YRD and GBA. JJJ performed worse than the YRD and GBA in all three city branding strategy scores, especially CM and CB. To some extent, CM and CB strategies are more sophisticated strategies than CP. This indicates that city branding strategies in the YRD and GBA are more mature than in JJJ. Compared with the YRD and GBA, JJJ does not yet have well-developed CM and CB strategies. Similar findings are also presented in the study of [de Jong et al. \(2018\)](#), which shows that the branding of cities in JJJ shows a larger gap between city branding practices and their real situations than

in the other two leading megacity regions.

Since the economic level of cities in the YRD region is higher than the national average, and the industrial clusters are more abundant in this area, these cities perform well in CM strategies. Compared with cities in the YRD, the cities in the GBA have better performance in terms of CB strategies. Hong Kong, Macao, Shenzhen, and Guangzhou position themselves as international cities, which means this region pays more attention to its international image. These cities also influence other neighbour cities in the region when it comes to CB strategies. To some extent, cities on the mainland have paid attention to CP strategies in past years. As economic development still remains the priority of the mainland government, the municipal governments have made more of an

effort to employ CM strategies. However, only international cities, megacities, and their surrounding cities have paid attention to CB strategies. Björner (2006) found that the branding of megacities in China was influenced by the trend of internationalisation and globalisation worldwide. Our research further finds that the cities surrounding megacities are also starting to pay more attention to CB strategies following the branding practices of megacities recently.

In terms of the spatial relationship between city branding strategy adoption in a cross-regional comparison, we found that the bottom five cities in the three strategies are all small cities, except Macao, which indicates that small cities encounter more challenges in city branding strategy adoption (Ma et al., 2021). Although small cities claim they are adopting CB strategies, they have not shown the same performance as megacities. The cities with the top five CP, CM and CB strategy scores are in the YRD and GBA, with the exception of Beijing and Tianjin. This indicates that cities perform well in city branding strategies in specific regions. Additionally, regarding city branding strategy adoption, Shenzhen has the smallest standard derivation of three city branding strategies. Compared with other megacities, it has stronger CP and CB strategies. The reason may be because it has more opportunities to learn from Hong Kong's CP and CB strategies than other cities.

Above all, cross-regional comparisons of city branding strategy adoption show spatial relationships. Cities face similar social and economic situations within the same regions and can more easily interact with other cities in those regions. This recalls the study of the Negev region in Israel, which illustrates that branding at the top of the familiarity ranking can influence other lesser known brands (Aharon & Alfasi, 2022). In a cross-regional comparison, results show that the leading cities in city branding strategies tend to be located in specific areas, such as the YRD or GBA regions.

The spatial analysis method shows that city branding strategy adoption and its influencing factors in the three megacity regions entail spatial relationships. Regarding influencing factors, JJJ, the YRD and the GBA tend to have slightly different influencing factors of CP, CM and CB strategies. Generally, cities with larger populations, higher economic development levels, or higher tertiary proportions in the industrial structure tend to score higher in city branding strategies. This indicates that developed cities have more capabilities and opportunities to apply city branding strategies and to experience urban transformation (Ma et al., 2021). However, the influencing factors of city branding strategy adoption can be different in various regions. It is necessary to explore the influencing factors in the local context. Additionally, as the GWR model can provide differentiated coefficients for explained variables, the range of coefficients of different cities in the three regions is tiny. This shows that cities in the same regions tend to have similar influential intensity. This result demonstrates that the city branding strategies in the same region tend to be similarly influenced by certain factors.

6. Conclusion

This research explores the spatial relationships between city branding strategy adoption in megacity regions. In a within regional comparison, the YRD, the GBA and JJJ have features that are apparent in their city branding strategies. In a cross-regional comparison, the city branding strategy scores of leading and lagging cities all fall within the same region. Using the spatial analysis method, the Moran *I* test also indicates the spatial correlation of city branding strategy adoption in three megacity regions. Regarding influencing factors, JJJ, the YRD and the GBA tend to have slightly different factors influencing CP, CM and CB strategies. We find that the economic situation and population scale tend to influence city branding strategies in three regions. In addition, the city branding strategies in the same regions are similarly influenced by certain factors.

This research contributes to the literature by examining the spatial relationship of city branding strategy adoption in megacity regions from the pattern and influencing factor perspectives. First, from a theoretical

perspective, previous studies discussed the geographical features of place branding (Acharya & Rahman, 2016; Aharon & Alfasi, 2022). This research further shows that city branding strategy adoption in megacity regions has spatial relationships, which means that cities interact with each other in their branding strategy adoption. Second, from a methodological perspective, previous studies showed spatial relationships in city branding by using limited study samples and qualitative research (Ma et al., 2021; Aharon & Alfasi, 2022). By adopting the Geographic Weighted Regression method, we provided fresh evidence of the spatial relationships in both city branding strategy adoption patterns and their influencing factors.

This research has several policy implications. First, as city branding strategy adoption has spatial relationships in the regional context, policymakers need to consider these spatial relationships when considering city branding design. Policymakers can use the branding of megacities to influence the city branding strategies of surrounding cities in megacity regions. Second, city branding strategies are influenced by different economic or urban factors, and policymakers need to pay attention to these factors in a regional context, such as economic development status and population scale. As the situation of Chinese megacity regions is likely similar to other regions in developing countries, the development stage of CP, CM and CB strategies can also provide lessons for regions like the Kolkata Metropolitan Area in India or the Southeast Region in Brazil. Third, as CP, CM and CB strategy can complement the spatial planning of megacity regions, policymakers can also tailor their own city branding strategy adoption to strengthen the implementation of spatial planning.

This research also has several limitations. First, we only examined the city branding strategy adoption in Chinese megacity regions, which are governed under the Chinese political and institutional framework. In China, local governments pay more attention to their planning documents and government work reports, which can be a reliable source for CM and CB strategies. In other countries, it can be challenging to find the same information from these standardized government documents. Future research can pay more attention to the urban reality of city branding strategies with different regional contexts and political systems, which can also contribute to existing city branding theories. Second, when city branding contents are co-created by government, citizens and other stakeholders, more factors should be considered, rather than only considering urban or economic factors. Future research can also study the interaction between city branding strategy from the government side and city branding co-creation from the other stakeholder side. Third, the data in this research are second-hand, and future research can collect more detailed information based on interviews, such as the influence of stakeholder involvement, higher-level government, city competition, or copycat behaviours of city branding strategy adoption. Regarding future research suggestions, it would be possible to examine the differences between what city branding strategy policymakers claim in city websites, the city branding activities on the ground and what residents recognise as the reality via questionnaires.

CRediT authorship contribution statement

Haiyan Lu: Conceptualization, Methodology, Data collection and curation, Formal analysis, Software, Visualization, Writing - original draft; Writing - review & editing;

Wenting Ma: Conceptualization, Methodology, Data collection and curation, Funding acquisition; Formal analysis, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was funded by the Fundamental Research Funds for the

Central Universities (Grant No. HIT.HSS.202220), the Project of Philosophy and Social Sciences Planning of the Guangdong Province (Grant No. GD22YGL09; GD22XGL54). This work was also grateful for the support from Shenzhen City Overseas High-Level Talents Introduction Funding and Shenzhen Humanities & Social Sciences Key Research Base for Carbon Emission Peaking and Carbon Neutral Technology, Policy, and Management, Harbin Institute of Technology, Shenzhen.

Appendix A

Table A1

Evaluation standard of indicators of City Promotion strategies.

Code	Indicators	5	4	3	2	1
CP1	Presence of a city logo	The city has a specially designed logo, which appears on the official government website	None	Apply some changed font to the logo on the official websites	None	The city doesn't have a special logo
CP2	Presence of a city slogan	The city has special slogans that can reflect the characteristics of the city	None	The city has its own slogans, but they do not reflect the characteristics of the city	None	The city doesn't have a special slogan.
CP3	Presence of a video of the city	The city has some city image videos. The videos are carefully crafted and often updated	None	The city has a special advertising campaign, but it is not updated very frequently	None	The city doesn't have a special city image video
CP4*	Presence of another media platform to promote the city in an indirect way	There are five ways to access the website, such as Weibo and WeChat, etc.	There are four ways to access the website, such as Weibo and WeChat, etc.	There are three ways to access the website, such as Weibo and WeChat, etc.	There are two ways to access the website, such as Weibo and WeChat, etc.	There is one way to access to the website, such as Weibo and WeChat, etc.

Note: "Abundant," "relatively large number," "some," "a few," and "no" are descriptions based on a range from the smallest to largest, which was divided into five scales.

Table A2

Evaluation standard of indicators of City Marketing strategies.

Code	Indicators	5	4	3	2	1
CM1-1	The city has constructed some tourism demonstration areas	There are abundant tourism demonstration areas	There are a relatively large number of tourism demonstration areas	There are some tourism demonstration areas	There are a few tourism demonstration areas	There is no tourism demonstration area
CM1-2	The city has the tourism projects	There are abundant tourism projects	There are a relatively large number of tourism projects	There are some tourism projects	There are a few tourism projects	There is no tourism project
CM1-3	The city hosts tourism activities	They hosted abundant tourism activities	They hosted a relatively large number of tourism activities	They hosted some tourism activities	They hosted a few tourism activities	They did not host any tourism activities
CM2-1	The city seeks to reform business systems and improve business services	They have abundant reform in the business system	They have a relatively large number of reforms in the business system	They have some reforms in the business system	They have a few reforms in the business system	They have no reforms in the business system
CM2-2	The city provides a special fund to attract companies	Abundant budget allocated	Relatively large budget allocated	Some budget allocated	A small budget allocated to attract business	No budget allocated to attract business
CM2-3	The city holds the investment promotion meetings	Held abundant influential, high-level business activities, fairs, trade shows	Held a relatively large number of business activities, fairs, trade shows	Held some trade fairs	Held a small number of small impact, small scale marketing activities	They did not organise marketing campaigns
CM2-4	The city reduces business costs and tax burden to companies	Reduce abundant tax and business costs	Reduce a relatively large number of tax and business costs	Reduce some tax and business costs	Reduce a few tax and business costs	No tax breaks
CM3-2	The city provides preferential policies to attract talent	Provide abundant preferential policies	Provide relatively large number of preferential policies	Provide some preferential policies	Provide a few preferential policies	No preferential policy
CM3-3	The city has plans for talent recruitment	The city has abundant professional plans about talent recruitment	The city has a large number of professional plans about talent recruitment	The city has some professional plans about talent recruitment	The city has a few professional plans about talent recruitment	The city doesn't provide any plans
CM3-4	The city improves a service level for talent	Adopt abundant measures to improve the service level for talent	Adopt a relatively large number of measures to improve the service level for talent	Adopt some measures to improve the service level for talent	Adopt a few measures to improve the service level for talent	There is no measure to improve the service level for talent
CM3-5	The city hosts talent recruitment activities and conferences	Held abundant activities to attract talent with high influence	Held a relatively large number of activities to attract talent with high influence	Held some marketing activities	Held a few small impact, small scale marketing activities	There was no marketing campaign

Note: "abundant," "relatively large number," "some," "a few," and "no" are descriptions based on a range from the smallest to largest, which was divided into five scales.

Table A3

Evaluation standard of indicators of City Branding strategies.

Code	Indicators	5	4	3	2	1
CB1	Presence of a professional website to do city branding	The city has a special city brand website, which updates information in time and constructs it well		The city has a special city brand website, but the website construction is not professional		The city doesn't have any special city branding websites
CB2	Presence of a specific city branding organisation or department	The city sets up a special city branding organisation or department, which is very professional		The city has a special city branding organisation or department		The city doesn't have any special city branding organisation or department
CB3	Presence of city branding documents identifying construction and implementation	The city releases abundant documents about city branding construction and implementation	The city releases a relatively large number of documents about city branding construction and implementation	The city releases some documents about city branding construction and implementation	The city releases a few documents about city branding construction and implementation	The city doesn't release any documents about city branding construction and implementation
CB4	Presence of a city brand identity and coherent goal, vision, strategy, roadmap, or policy in place	The city has a clear vision of urban development, in line with the reality of the city. The identity of the city combines with the city's characteristics, resources, and conditions. The city identity has a very high differentiation	The city has a clear vision of urban development. The identity of the city can be distinguished by other cities	The characteristics of a city can be seen from the nature of the city	The degree of differentiation is not high	The city doesn't have a clear identity and it doesn't combine with the city's characteristics
CB5	The city regularly organises professional city branding activities, including conferences, forums, or mega events	The city hosts abundant influential international and domestic events	The city hosts a relatively large number of international and national events	The city has hosted some international, national, and provincial level events	The city hosted a few international, national, and provincial and municipal events	The city hosts very few events

Note: "abundant," "relatively large number," "some," "a few," and "no" are descriptions based on a range from the smallest to largest, which was divided into five scales.

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